

Phase 3 - Language Definition

4 Knowledge Layer, Schemas, Ontologies

The cleaned datasets contain a range of entities and properties that are essential for analyzing oncoviruses and the interactions between viral proteins and host factors that disrupt cellular pathways leading to cancer. For most entity types, no suitable schema from Schema.org or Bioschemas.org exists, so we rely on the schema derived directly from the datasets. However, when applicable, certain entity types are aligned with established standards.

4.1 Cancer (*from the original data*)

- **ID** (Integer, from the original data): Unique identifier for the entry.
- **Type** (Text, from the original data): Category or type of cancer.
- **PrimarySite** (Text, from the original data): Anatomical site or primary location associated with the cancer type.

4.2 HostProtein (*from the original data*)

- **ID** (Integer, from the original data): Unique identifier for the entry.
- **HostProteinID** (Text, from the original data): Reference to the UniProt Accession Number of the HostProtein.
- **HostGeneName** (Text, from the original data): Gene name of the host protein.
- **HostProteinName** (Text, from the original data): Protein name of the host protein.

4.3 ViralProtein (*from the original data*)

- **ID** (Integer, from the original data): Unique identifier for the entry.
- **ViralProteinID** (Text, from the original data): Reference to the UniProt Accession Number of the ViralProtein.
- **ViralGeneName** (Text, from the original data): Gene name encoding the viral protein.
- **ViralProteinName** (Text, from the original data): Protein name of the viral protein.

4.4 ProteinAnnotation (*from the original data*)

- **ID** (Integer, from the original data): Unique identifier for the entry.
- **ProteinID** (Text, from the original data): Protein's UniProt Accession Number.

- **Function** (Text, from the original data): Functional role of the protein.
- **Localization** (Text, from the original data): Protein's subcellular localization.

4.5 Vaccine *(from the original data)*

- **ID** (Integer, from the original data): Unique identifier for the entry.
- **Name** (Text, from the schema.org): Name of the Vaccine.
- **Description** (Text, from schema.org): Description of the Vaccine.
- **Source** (URL, from the original data): A reference to the origin of the information, typically a webpage, database entry, report, or publication from which the Vaccine's entity data was obtained.

4.6 Virus *(from schema.org)*

- **ID** (Integer, from the original data): Unique identifier for the entry.
- **VirusCode** (Text, from the original data): Code for the virus.
- **OrganismName** (Text, from the original data): Scientific or common name of the virus.
- **Species** (Text, from the original data): Viral species name.
- **Host** (Text, from the original data): Primary host organism.

4.7 Phenotype *(from bioschemas.org)*

- **ID** (Integer, from the original data): Unique identifier for the entry.
- **Type** (Text, from the original data): Category or type of the virus.
- **Source** (URL, from the original data): A reference to the origin of the information, typically a webpage, database entry, report, or publication from which the Virus's entity data was obtained.
- **Trait** (Text, from the original data): Described phenotypic trait.

4.8 Pathway *(from the original data)*

- **ID** (Integer, from the original data): Unique identifier for the entry.
- **PathwayID** (Text, from the original data): Cross-reference to pathway KEGG database
- **PathwayName** (Text, from the original data): Name of the biological pathway.
- **Category** (Text, from the schema.org): Pathway classification.
- **Description** (Text, from Schema.org): Description of the pathway.
- **Source** (URL, from the original data): A reference to the origin of the information, typically a webpage, database entry, report, or publication from which the pathway's entity data was obtained.

This schema captures the structure of the datasets while integrating Schema.org and Bioschemas.org standards when suitable, offering a comprehensive framework for

analyzing oncoviruses and the interactions between viral proteins and host proteins that disrupt cellular pathways leading to cancer.

5 Language Definition

The primary goal of the language definition phase is to formally specify the concepts and information that will be included in the final Knowledge Graph to fulfill the objectives of this project. To achieve this, we first identified all relevant concept structures, including E-Types, relationships, and object properties, that needed formal specification. Each identified concept was then aligned with existing ontologies, first searching on the Universal Knowledge Core (UKC) to examine whether the concept was already defined. If a concept was not present, it was subsequently checked against additional specialized resources, and if still not found, it was defined as a custom concept specific to our project. Due to the nature of this study, many of the elements are highly domain-specific, for example, the E-Type HostProtein and its properties, and therefore required a complete formal definition within our framework. Since our dataset contains numerous concepts specific to molecular biology, virology, and immunology, we relied on specialized ontologies to retrieve definitions and standard identifiers, completing our Language Resource Table. In particular, we used ontologies from NCBO BioPortal.

5.1 Concept Identification

We produced the Language Resource Table, in which each identified concept is listed and organized into three columns:

- Column 1: ConceptID: This takes one of three formats depending on where the concept was found: a UKC ID (ID: UKC-number) if present in the Universal Knowledge Core, a link to an external ontology if found elsewhere, or a custom project ID (ID: OKG25-number) if defined internally.
- Column 2: Word_en: The name or word(s) associated with the concept.
- Column 3: Gloss-en: A clear explanation or definition for each concept.

In order to improve readability, we applied the following color-coding scheme:

- **Blue:** Represents ETypes from the ER Model
- **Green:** Represents Attributes from the ER Model
- **Red:** Represents Relationships from the ER Model

ConceptID	Word-en	Gloss-en
UKC-6543	Virus	Ultramicroscopic infectious agent that replicates itself only within cells of living hosts; many are pathogenic; a piece of nucleic acid (DNA or RNA) wrapped in a thin coat of protein.
UKC-24644	Vaccine	Immunogen consisting of a suspension of weakened or dead pathogenic cells injected in order to stimulate the production of antibodies.
UKC-26778	Phenotype	What an organism looks like as a consequence of the interaction of its genotype and the environment.
UKC-67961	Cancer	Any malignant growth or tumor caused by abnormal and uncontrolled cell division; it may spread to other parts of the body through the lymphatic system or the blood stream.
http://purl.obolibrary.org/obo/COMO_0000067	ProteinAnnotation	Additional information and metadata on proteins.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C17254	ViralProtein	Proteins encoded by viral genes.
http://purl.obolibrary.org/obo/IDOMAL_0001068	HostProtein	A protein that is produced by a host.
http://www.biopax.org/release/biopax-level3.owl#Pathway	Pathway	A set or series of interactions, often forming a network, which biologists have found useful to group together for organizational, historic, biophysical or other reasons. Usage: Pathways can be used for demarcating any subnetwork of a BioPAX model. It is also possible to define a pathway without specifying the interactions within the pathway. In this case, the pathway instance could consist simply of a name and could be treated as a 'black box'. Pathways can overlap, i.e. a single interaction might belong to multiple pathways. Pathways can also contain sub-pathways. Synonyms: network, module, cascade, Examples: glycolysis, valine biosynthesis, EGFR signaling.
UKC-2	Name	A language unit by which a person or thing is known.
UKC-695	Localization	A determination of the place where something is.
UKC-6810	Host	An animal or plant that nourishes and supports a parasite; it does not benefit and is often harmed by the association.
UKC-31356	Category	A general concept that marks divisions or coordinations in a conceptual scheme.
UKC-31362	Type	A subdivision of a particular kind of thing.
UKC-105757	Function	Serve a purpose, role, or function.
http://hl7.org/fhir/id	ID	Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.

http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C25365	Description	A written or verbal account, representation, statement, or explanation of something.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C25683	Source	Where something is available or from where it originates.
http://edamontology.org/data_2909	OrganismName	The name of an organism (or group of organisms).
http://edamontology.org/data_1045	Species	The name of a species (typically a taxonomic group) of organism.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C85496	Trait	Any genetically determined characteristic.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C164895	PathwayName	Name of the biological pathway.
http://sbmi.uth.tmc.edu/ontology/ochv#C0449695	PrimarySite	Primary Site / Anatomical site or primary location associated with the cancer type.
OKG25-01	VirusCode	Unique identifier assigned to a virus.
OKG25-02	PathwayID	KEGG database identifier referencing a specific biological pathway.
OKG25-03	ProteinID	Unique Uniprot Reference to the protein.
OKG25-04	ViralProteinID	Unique Uniprot Reference to the viral protein.
OKG25-05	HostProteinID	Unique Uniprot Reference to the host protein.
OKG25-06	ViralGeneName	Standard name assigned to a viral gene.
OKG25-07	HostGeneName	Standard name assigned to a host gene.
OKG25-08	ViralProteinName	Official or common name of a viral protein.
OKG25-09	HostProteinName	Official or common name of a host protein.

UKC-98182	targets	Intend (something) to move towards a certain goal.
UKC-100755	causes	Give rise to; cause to happen or occur, not always intentionally.
UKC-103527	has	Have or possess, either in a concrete or an abstract sense.
OKG25-10	encodes	Represents that a genetic element (such as a gene, transcript, or genomic region) of an organism specifies the sequence or structure of a functional product, typically a protein or RNA molecule. The relationship links the genetic source (subject) to the resulting biological product (object).
OKG25-11	involvedIn	Indicates that an entity participates in, contributes to, or plays a role within a process, event, pathway, or higher-level activity. The relationship is generally directional, linking the participating entity (subject) to the broader process or context (object).
OKG25-12	interactsWith	Denotes that two entities engage in a direct or measurable interaction, influencing each other's state, function, or behavior. This relationship is typically symmetric, reflecting mutual participation in the interaction.